

DE MONTFORT UNIVERSITY, LEICESTER

Faculty of Computing
MSc Artificial Intelligence – Project

Trustworthy RAG Assistant with Confidence Scoring

A Grounded Medical Question-answering system with explainable per-answer Confidence, Honest refusal and Cited evidence

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Declaration

I declare that this project report is my own work and that the work reported in it has been carried out by me under the guidance of my supervisor **Dr. Mario A. Gongora**, except where explicit reference is made to the contribution of others. All sources of information have been acknowledged by means of references, and the work has not been submitted, in whole or in part, for any other academic award. Where artificial-intelligence tools were used to support the development of the software and the preparation of this document, they were used in accordance with university guidance, and I remain fully responsible for the content, correctness and integrity of the work presented.

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Abstract

Large language models (LLMs) are increasingly consulted for medical information, yet they hallucinate producing fluent but incorrect statements and standard interfaces expose no signal of when an answer should be trusted. This project designs, implements and evaluates a trustworthy medical question-answering assistant that combines Retrieval-Augmented Generation (RAG) over the MedQuAD Data of National Institutes of Health (NIH) question answer pairs with an explainable, per answer confidence score. The system indexes 16,357 cleaned question answer pairs as 69,515 search keys in a FAISS vector store using all-MiniLM-L6-v2 sentence embeddings, retrieves complete parent answers through question-to-question semantic matching, and generates answers with an LLM accessed through the Groq API under a strict grounding prompt. Confidence is computed as a weighted combination of three measured signals self-consistency agreement between repeated generations, retrieval evidence strength, and answer groundedness and every answer is delivered with cited sources, a low-confidence warning below a 60% threshold, and an honest refusal when the corpus contains no relevant evidence. Evaluation on held-out MedQuAD questions found that 8 of 12 answerable questions were answered correctly under a semantic-similarity criterion, that mean confidence was 15.8 percentage points higher for correct answers (95.9%) than for incorrect ones (80.1%), and that all six deliberately unanswerable questions were correctly refused. A comparison against verbalized confidence reproduced the overconfidence reported in prior work. The contribution is an integrated, reproducible, zero-cost reference implementation demonstrating that model-agnostic confidence signals can meaningfully separate reliable from unreliable answers, together with an honest analysis of calibration limitations on a small evaluation set.

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List of Abbreviations

List of abbreviations used in this report.

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma-Separated Values
FAISS	Facebook AI Similarity Search
GPU	Graphics Processing Unit
GUI	Graphical User Interface

Abbreviation	Meaning
LLM	Large Language Model
NIH	National Institutes of Health (United States)
NLP	Natural Language Processing
PCA	Principal Component Analysis
Q&A	Question and Answer
RAG	Retrieval-Augmented Generation
RQ	Research Question
TPD	Tokens Per Day (API quota)
UI	User Interface

Chapter 1 – Introduction:

1.1 Background

Generative artificial intelligence, and in particular large language models (LLMs) built on the Transformer [1] architecture, has changed how people search for information. Models trained at scale acquire striking few-shot abilities and produce answers that read as fluent, authoritative prose. This fluency, however, is produced by next token prediction: at no point does the training objective verify that a generated statement is true. The resulting failure mode fluent but false output, known as **hallucination** is now recognised as one of the principal barriers to the safe deployment of LLMs.

The consequences of hallucination depend on the domain. In casual use an incorrect answer is an inconvenience; in medicine it is a hazard. Members of the public increasingly ask AI assistants health questions, and empirical work has shown that even clinically oriented LLMs produce a meaningful fraction of answers that expert reviewers judge potentially harmful. Retrieval-Augmented Generation (RAG) mitigates part of the problem: instead of answering from parametric memory, the model first retrieves passages from a trusted corpus and answers from that evidence, which improves factuality and enables source attribution. Yet retrieval is not a cure. Surveys of the field report that generators can ignore retrieved context, answer from memory, or cite passages that do not support the claim. Grounding therefore needs to be paired with a second mechanism: an estimate of how reliable each individual answer is, communicated to the user. This project builds and evaluates exactly that combination.

1.2 Problem Statement

A standard LLM response is insufficient for trustworthy question answering for three reasons. First, the model may hallucinate, and its confident tone gives the user no basis for doubt. Second, even when retrieval grounding is added, the user cannot verify an answer without access to the supporting evidence. Third, and most fundamentally, neither a plain LLM nor a plain RAG system tells the user “*How much to trust this specific answer*”: modern neural systems are systematically overconfident, and simply asking a model to state its confidence produces scores clustered near the top of the scale regardless of accuracy. The problem addressed here is therefore the absence of an integrated, accessible system that grounds medical answers in trusted evidence, attaches a measured and explainable confidence signal to every answer, warns the user when that signal is low, and refuses honestly when the evidence does not exist.

1.3 Motivation

The motivation for this project is both practical and academic. Practically, consumer-facing medical AI is being adopted faster than its reliability can be assured, and low-cost, transparent safety mechanisms are needed for the settings students, small clinics, low-resource organizations, where commercial guardrails are unaffordable. Academically, grounding (the RAG literature) and uncertainty estimation (the confidence literature) have developed largely in isolation: retrieval papers measure accuracy but rarely surface uncertainty to users, while confidence papers are evaluated on detection benchmarks rather than inside deployed, user-

facing systems. Building one working system that combines both, under realistic free-tier constraints, produces evidence that neither literature currently supplies.

1.4 Research Aim

Aim: To design, implement and evaluate a free, end-to-end medical question-answering assistant that grounds every answer in the MedQuAD corpus through retrieval-augmented generation, attaches an explainable per-answer confidence score derived from measured system signals, warns the user when confidence is low, refuses when the corpus contains no relevant evidence, and presents the supporting sources for verification.

1.5 Research Objectives

- **O1** - Acquire, clean and index the MedQuAD medical question-answer corpus into a semantic vector store suitable for retrieval.
- **O2** - Implement a RAG pipeline in which answers are generated by an API-hosted LLM strictly from retrieved corpus content.
- **O3** - Design and implement a per-answer confidence score combining self-consistency agreement, retrieval evidence strength and answer groundedness.
- **O4** - Implement trust controls: a low-confidence warning threshold, honest refusal for out-of-corpus questions, and per-answer source citation.
- **O5** - Develop a professional graphical user interface exposing the answer, confidence breakdown, warnings and evidence to end users.
- **O6** - Evaluate answer quality, confidence behavior, refusal accuracy and calibration on held-out data, and compare the confidence method against a verbalised-confidence baseline.
- **O7** - Critically analyse failures and limitations, and identify improvements.

1.6 Research Questions

The project answers four research questions, refined from the approved proposal so that each is answerable from the experiments actually performed:

- **RQ1** - Can retrieval-augmented generation over a trusted medical corpus produce grounded answers to consumer health questions using only free tools and a free API-hosted LLM?
- **RQ2** - Can a model-agnostic confidence score, computed from answer agreement, retrieval strength and groundedness, meaningfully separate correct from incorrect answers?
- **RQ3** - Does the implemented system refuse honestly when questions fall outside its knowledge base, rather than fabricating answers?
- **RQ4** - How does the proposed confidence method compare with verbalised confidence, and what are its calibration limitations?

1.7 Scope

The project covers English-language medical question answering over the MedQuAD corpus; a complete RAG pipeline (embedding, vector indexing, retrieval, grounded generation); the three-component confidence mechanism with warning and refusal behaviour; a Streamlit web interface; and a quantitative and qualitative evaluation. It deliberately excludes personalized medical advice or diagnosis (the system is an information aid only), the training or fine-tuning of language models (all models are used as released), non-English content, clinical or patient data of any kind, and integration with healthcare systems. These exclusions bound the ethical exposure of the work and are revisited in Chapter 7.

1.8 Contributions

- **C1 - An integrated trustworthy-RAG reference implementation.** A complete, working system that unifies grounding, explainable confidence, warnings, refusal and citation, built entirely from free components and reproducible by any student.
- **C2 - A three-component, model-agnostic confidence score.** A per-answer score combining self-consistency, retrieval evidence and groundedness, requiring no access to model internals and therefore portable across API providers.
- **C3 - Empirical evidence on confidence behaviour.** An evaluation showing a 15.8-percentage-point mean confidence separation between correct and incorrect answers, perfect refusal on out-of-corpus questions, and a reproduction of verbalised-confidence overconfidence, together with an honest calibration analysis on a small sample.
- **C4 - An engineering account of free-tier constraints.** Documented solutions for API token quotas, model deprecation and session-persistence in Google Colab, of practical value to resource-constrained developers.

1.9 Report Structure

Chapter 2 critically reviews the literature on LLMs, hallucination, RAG, confidence estimation and medical question answering, and derives the research gap. Chapter 3 details the methodology: dataset, preprocessing, retrieval design, generation, confidence scoring and the GUI. Chapter 4 describes the system architecture and implementation with code extracts and interface figures. Chapter 5 presents the experimental setup and results. Chapter 6 discusses the results critically, analyses failures and answers the research questions. Chapter 7 examines limitations, ethics, reproducibility and future work, and Chapter 8 concludes.

Chapter 2 - Literature Review:

This chapter reviews the research on which the project is built. It is organized from the cause of the problem (How LLMs generate text and why they hallucinate) through the two partial solutions the field has produced (retrieval grounding and confidence estimation) to the specific application domain (medical question answering) and its evaluation. Each section closes with a critical assessment, and Section 2.13 consolidates the research gap that the project addresses. The review deliberately concentrates on a focused set of key works that directly justify design decisions, treating each critically rather than descriptively.

2.1 Large Language Models

Modern LLMs derive from the Transformer, whose self-attention mechanism made it practical to train very large models on very large corpora. Brown et al. [2] demonstrated with GPT-3 that scale alone unlocks few-shot, in-context learning, establishing the decoder-only, autoregressive paradigm that underpins today's assistants, including the API-hosted model used in this project. The critical observation for this work is architectural: an autoregressive LLM is trained to predict the next token given preceding text. Nothing in this objective distinguishes true statements from plausible ones, so factual reliability is an emergent, unguaranteed property. Scale improves knowledge coverage but does not confer calibrated self-knowledge, which motivates every subsequent section of this review.

2.2 Hallucination in Generative AI

Ji et al. [3] provide the most widely cited survey of hallucination in natural language generation, distinguishing *Intrinsic* hallucination (output that contradicts the source material) from *Extrinsic* hallucination (output that cannot be verified against any source). Huang et al. [4] update the taxonomy for the LLM era, separating factuality errors from faithfulness errors, and catalogue causes including noisy training data, exposure bias and sampling-based decoding. Both surveys are strong on causes and general-purpose mitigations, but neither tells a developer how to build a safe system in one specific high-stakes domain: the mitigations they list reduce the rate of hallucination without guaranteeing any individual answer. For a medical assistant this distinction is decisive a user reading a single answer about their own health cares about that answer, not the average error rate and it is why this project treats per-answer confidence estimation as a first-class requirement rather than an optional extra.

2.3 Retrieval-Augmented Generation

Lewis et al. [5] introduced RAG, coupling a dense retriever with a sequence-to-sequence generator so that answers are produced from retrieved evidence rather than parametric memory alone, improving factuality on knowledge-intensive tasks and crucially for this project enabling provenance, since retrieved passages can be shown to the user. Their original design trained retriever and generator jointly, which is infeasible on a student budget; the field has since converged on the frozen-LLM, prompt-time pattern that Gao et al. [6] call **naive RAG**, in which documents are embedded offline, retrieved at query time and injected into the prompt of an unmodified LLM. Gao et al. [6] survey also catalogues advanced

extensions query rewriting, re-ranking, iterative retrieval but reports mostly incremental gains on specific benchmarks, each adding latency and API cost. Read critically, this supports the design choice made here: a clean prompt-time pipeline with strong retrieval is more defensible for a free-tier project than a complex pipeline that could not be properly ablated. The survey is equally clear that RAG reduces but does not eliminate hallucination, which motivates the confidence layer of Chapter 3.

2.4 Information Retrieval, Embeddings and Semantic Search

Retrieval quality determines everything downstream. Karpukhin et al. [7] showed with Dense Passage Retrieval that learned dense representations of questions and passages substantially outperform sparse term-matching on open-domain question answering, establishing dense bi-encoders as the default retrieval architecture. Reimers and Gurevych [8] made this efficient at scale: Sentence-BERT fine-tunes Transformer encoders in a siamese configuration so that a single vector per sentence supports cosine-similarity search, reducing comparison costs from hours to milliseconds. The embedding model used in this project, all-MiniLM-L6-v2, is a distilled member of this family 384 dimensions, approximately 80 MB chosen for its strong quality-to-cost ratio on free hardware. The acknowledged trade-off is that it is a general-purpose encoder rather than a biomedical one; a domain-tuned embedder would likely retrieve better but at a cost in size, and the effect of this choice is visible in the retrieval analysis of Chapter 6.

2.5 Vector Similarity Search

Efficient nearest neighbour search over tens of thousands of vectors is provided by FAISS [9], which offers exact and approximate index structures with GPU acceleration. At the scale of this project (69,515 vectors), an exact index suffices, and FAISS's serialization supports the offline-build, online-load pattern used here: the index is built once in Google Colab and persisted to Google Drive, so later sessions load it in seconds. The critical point is that vector databases are infrastructure, not intelligence: FAISS returns the nearest vectors to a query under the geometry the embedding model created, so retrieval failures analyzed in Chapter 6 are properties of the embedding space and the query formulation, not of the index.

2.6 Trustworthy AI

Ni et al. [15] survey trustworthiness in RAG specifically and argue that it spans multiple dimension's reliability, robustness, transparency, accountability rather than accuracy alone, identifying honest abstention and verifiable source attribution as open needs in deployed systems. As a recent preprint the survey maps the field rather than settling it, but its framing matches this project closely: the implemented combination of grounded answers, per-answer confidence, low-confidence warnings, refusal and visible sources operationalises precisely the properties the survey identifies as missing. Transparency here is pragmatic rather than formal: showing the retrieved evidence lets the user verify claims directly, avoiding the faithfulness problems of post-hoc explanation methods, at the cost of explaining only what evidence was available, not how the model weighed it.

2.7 Confidence Estimation in LLMs

The classical foundation is calibration. Guo et al. [10] demonstrated that modern neural networks are systematically overconfident reported probabilities exceed empirical accuracy introduced the Expected Calibration Error, and showed that simple post-hoc temperature scaling helps. Their methods, however, require access to model logits, which hosted APIs such as the one used here do not expose; the calibration literature therefore defines this project's **evaluation standard** (confidence should match accuracy, examined in Section 5.8) but cannot supply its method. A second line asks the model to assess itself: Kadavath et al. [19] found that large models can partially predict whether their own answers are true, but demonstrated this with internal model access. The fully black-box version prompting the model to state a confidence percentage was tested systematically by Xiong et al. [14] who found verbalised confidence clustered in the 80–100% range regardless of accuracy across multiple models. Their conclusion that consistency-based estimates outperform verbalised ones is a key empirical justification for the method adopted here, and Chapter 5 independently reproduces the overconfidence finding.

2.8 Self-Consistency and Sampling-Based Confidence

Wang et al. [11] proposed self-consistency for accuracy: sample multiple reasoning paths and take the majority answer. The degree of agreement between samples is itself an uncertainty signal, and Manakul et al. [12] exploit exactly this in SelfCheckGPT: if a model knows a fact, stochastic samples agree on it; if it is guessing, they contradict each other. SelfCheckGPT requires no logits, no training data and no external database, making it one of the few methods deployable against any API. Kuhn et al. [13] refine the idea into semantic uncertainty, clustering samples by meaning (via entailment) before measuring dispersion, so that paraphrases count as agreement; the approach has the strongest published evidence but requires an additional entailment model and more samples per question. This project adopts self-consistency as the core signal, measures agreement with embedding similarity rather than exact matching borrowing Kuhn et al. [13] central insight in a lightweight form and augments it with two signals the RAG setting makes available for free: retrieval evidence strength and answer groundedness. The known weakness, that a consistently wrong model still scores high, is examined empirically in Chapter 6.

2.9 Medical Question Answering

The application domain supplies both the motivation and the data. Ben Abacha and Demner-Fushman [16] created MedQuAD, question-answer pairs curated from twelve trusted NIH websites covering diseases, drugs and procedures; although their paper concerns question entailment, the corpus is the lasting contribution and is unusually well suited to a trustworthy assistant because its answers originate from authoritative sources. On the model side, Singhal et al. [17] provide the strongest evidence of what LLMs can and cannot do in medicine: Med-PaLM reached 67.6% on US medical licensing questions, but expert review judged model answers clearly inferior to clinicians', with a meaningful fraction rated potentially harmful. Two lessons follow: raw LLM answers are not safe enough for medical use, supporting grounding in a trusted corpus; and expert human review is too slow and expensive to apply per answer, which is precisely the role an automatic per-answer confidence signal fill.

2.10 Evaluating RAG Systems

Evaluating free-text answers is itself unsolved. Es et al. [18] propose RAGAS, reference-free faithfulness and relevance metrics computed with an LLM judge; the approach is attractive when gold answers are unavailable but circular, since the judge inherits its own biases. This project instead exploits the structure of its corpus: held-out MedQuAD pairs supply real gold answers, so answer quality is measured as embedding similarity between the generated and gold answers, with a threshold defining a correctness proxy, complemented by qualitative inspection and a calibration-style analysis in the spirit of Guo et al. [10]. The limitations of this proxy semantic similarity are not clinical correctness are acknowledged explicitly in Chapter 7.

2.11 Existing RAG Systems

Beyond component research, end-to-end medical RAG has been benchmarked at scale: work aggregated around toolkits and benchmarks shows retrieval lifting medical QA accuracy substantially across corpora and models. Yet published systems overwhelmingly report accuracy on multiple-choice examinations, stop short of exposing uncertainty to end users, and rarely implement abstention. Deployed commercial assistants, conversely, present answers with uniform authority regardless of evidential support. Between benchmark research and commercial products there is a shortage of open, inspectable systems that demonstrate the full trust stack grounding, confidence, warning, refusal, citation in one place, which is the niche this project occupies.

2.12 Critical Comparison

Table 2.1 - Critical comparison of prior approaches relevant to this project.

Work	Approach	Domain / setting	Strength	Limitation	Relevance here
Lewis et al. [5]	Joint dense retriever + generator	Open-domain QA	Established grounding; provenance	Joint training infeasible at low cost; no uncertainty	Grounding foundation
Gao et al. [6]	Survey: naive → modular RAG	General	Maps design space	Advanced additions costly, small gains	Justifies prompt-time design
Guo et al. [10]	Post-hoc calibration (logits)	Classification	Defines calibration & ECE	Needs white-box access	Evaluation standard only
Xiong et al. [14]	Verbalised confidence study	Multiple LLMs	Systematic empirical test	Shows method is overconfident	Baseline reproduced in Ch.5
Wang et al. [11]	Self-consistency sampling	Reasoning tasks	Model-agnostic; black-box	Targets accuracy, not user-facing confidence	Core confidence signal
Manakul et al. [12]	SelfCheckGPT consistency	Hallucination detection	Zero-resource, API-compatible	Detection benchmark, not deployed system	Method basis
Kuhn et al. [13]	Semantic uncertainty	Free-text generation	Meaning-aware; strong evidence	Extra NLI model + samples; costly	Lightweight variant adopted
Singhal et al. [17]	Med-PaLM + expert review	Medical QA	Clinical evaluation framework	Closed model; expert review unscalable	Motivates per-answer signal

Work	Approach	Domain / setting	Strength	Limitation	Relevance here
Ben Abacha & Demner-Fushman	MedQuAD corpus	Consumer health QA	Authoritative NIH sources	2019 snapshot; partial public copies	Knowledge base used

2.13 Research Gap

Three gaps emerge from this review. First, **Integration**: grounding and confidence estimation are mature separately but rarely combined in a working, user-facing system that also implements warning, refusal and citation - the trust stack that surveys identify as missing. Second, **Access Realism**: much confidence research assumes white-box model access, whereas real developers face black-box APIs; there is little published evidence on what can be achieved under free-tier, API-only constraints. Third, **Domain and Format**: medical RAG evaluation is dominated by multiple-choice examinations, while the public asks free-text consumer questions of exactly the kind MedQuAD contains. This project addresses all three: it builds the integrated system (gap one) under genuinely free, black-box constraints (gap two), and evaluates it on held-out free-text consumer health questions (gap three).

2.14 Summary

The literature establishes that hallucination is inherent to autoregressive generation, that retrieval grounding mitigates but does not eliminate it, that dense embedding retrieval and FAISS provide the practical machinery, and that among confidence methods only sampling-based approaches are simultaneously effective and deployable against black-box APIs. The medical literature justifies both the corpus and the need for per-answer reliability signals. The next chapter turns these findings into a concrete methodology.

Chapter 3 – Methodology:

3.1 Overall Research Methodology

The project follows a constructive (design-science) methodology combined with quantitative evaluation: an artefact the trustworthy RAG assistant is designed from the findings of the literature review, implemented iteratively, and then evaluated against measurable criteria derived from the research questions. Development proceeded in increments, each validated before the next was added: data preparation and indexing; baseline RAG; the confidence layer; trust controls (warning and refusal); the GUI; and finally, the evaluation harness. Two significant design revisions occurred during development and are reported honestly in Section 4.9: a retrieval redesign after fragment-level search proved unreliable, and token-efficiency changes after free-tier quota limits were reached. All development took place in Google Colab.

3.2 System Requirements

Functional requirements. The system shall: (FR1) answer free-text medical questions using only content retrieved from the MedQuAD corpus; (FR2) attach a confidence percentage to every answer, decomposed into its component signals; (FR3) display a prominent warning when confidence falls below a threshold; (FR4) refuse explicitly when no relevant corpus content exists, rather than guessing; (FR5) display the retrieved source passages with relevance scores for every answer; and (FR6) provide these functions through both a notebook interface and a web GUI.

Non-functional requirements. The system shall: (NFR1) cost nothing to build, run and host; (NFR2) respond to a question in interactive time on free infrastructure; (NFR3) avoid storing user questions or any personal data; (NFR4) keep API credentials out of source code; and (NFR5) be reproducible by a third party from the notebook and this report.

3.3 Dataset

The Dataset is MedQuAD, a corpus of consumer-health question–answer pairs curated from twelve NIH websites. The original corpus release contains 47,457 pairs; the publicly available Kaggle redistribution used in this project (medquad.csv, 22.3 MB) contains **16,412 rows**, reflecting the removal of certain sub-collections from public copies for copyright reasons a discrepancy verified during data inspection and important for reproducibility. Each row provides the question text, the answer text, the source website and a focus area (the medical entity concerned). Table 3.1 summarizes the corpus as actually used, and Figures 3.2–3.4 profile it.

Table 3.1 - Characteristics of the MedQuAD corpus as used in this project.

Characteristic	Value (measured from the data)
Source file	medquad.csv (Kaggle)
Rows in raw file	16,412 question–answer pairs
Rows after cleaning	16,357 pairs (55 removed)
Fields	question, answer, source, focus area

Characteristic	Value (measured from the data)
Mean question length	50 characters
Mean answer length	1,218 characters
Answer provenance	Official NIH websites
Access	Publicly available via Kaggle

3.4 Data Preprocessing

Preprocessing was deliberately conservative, since the answers originate from authoritative sources and aggressive normalization risks damaging clinical content. The implemented steps, in order, were: (i) automatic detection of the question and answer columns by name matching, making the loader robust to column-order changes in redistributions; (ii) removal of rows with missing questions or answers; (iii) whitespace normalization; (iv) removal of answers shorter than 30 characters and questions shorter than 10 characters, which inspection showed to be truncation artefacts; (v) removal of exact duplicate question-answer pairs; and (vi) truncation of extreme answers at 6,000 characters. These steps reduced the corpus from 16,412 to 16,357 rows, a deliberately small reduction (0.3%) consistent with a mostly clean source. Finally, 100 rows were withheld from indexing as an evaluation hold-out set (Section 5.3), leaving **16,257 pairs** in the knowledge base. Figure 3.1 summarizes the pipeline.

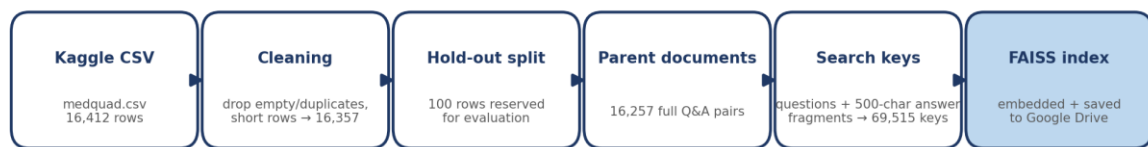


Figure 3.1 - Data preparation pipeline from the Raw Kaggle CSV to the saved FAISS index.

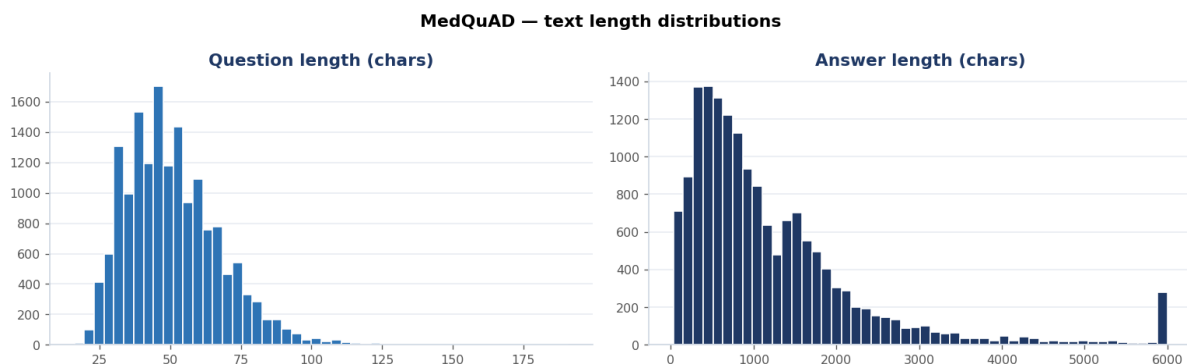


Figure 3.2 - Distribution of question-and-answer lengths in the cleaned MedQuAD corpus.

Figure 3.2 exposes the asymmetry that shaped the retrieval design: questions are short (mean 50 characters) while answers are long (mean 1,218 characters). Figure 3.3 shows the corpus is dominated by disease-oriented vocabulary (symptoms, treatments, causes), confirming its fit to consumer health questions, and Figure 3.4 shows the distribution of NIH source collections, which is retained as citation metadata.

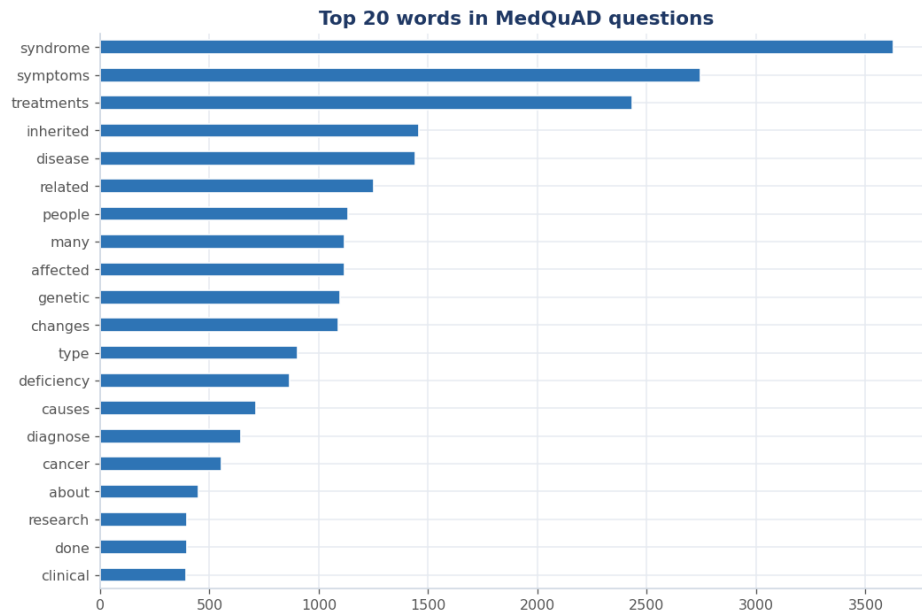


Figure 3.3 - Twenty most frequent content words in MedQuAD questions.

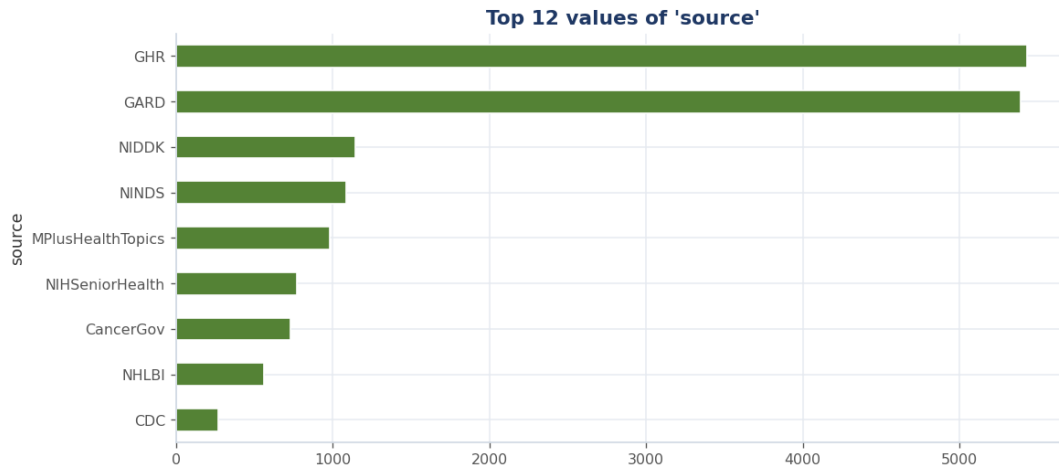


Figure 3.4 - Distribution of the twelve most frequent answer sources (focus areas) in the corpus.

3.5 Retrieval Units and Chunking

A central methodological decision, reached after the failure analysis described in Section 4.9, is the separation of search keys from delivery units. Each cleaned pair becomes a **parent document** holding the complete question, answer and source. Two kinds of search key are embedded, each tagged with its parent identifier: the dataset question itself (prefixed "Question: "), because a user question resembles a stored question far more than it resembles a fragment from the middle of an answer; and 500-character fragments of the answer with 50-character overlap, produced by a recursive character splitter, as a fallback for user questions phrased unlike any stored question. This yielded **69,515 search keys (16,257 question keys + 53,258 answer fragments)** for 16,257 parents. At answer time the model always receives complete parent answers, never fragments a parent-document strategy that preserves retrieval precision without sacrificing context completeness.

3.6 Embedding Generation

All search keys, queries and confidence computations use the same encoder: **all-MiniLM-L6-v2**, a distilled Sentence-BERT model producing 384-dimensional vectors, executed with normalized embeddings so that inner product equals cosine similarity. The model was chosen for its documented quality-to-cost ratio: at roughly 80 MB it embeds the full key set on a free Colab T4 GPU in under two minutes and runs acceptably on CPU in the deployed GUI, satisfying NFR1–NFR2. Using one encoder throughout keeps retrieval scores, agreement scores and groundedness scores in a single comparable geometry.

3.7 Vector Storage and Indexing

Vectors are stored in a FAISS index built once offline: the measured build over 69,515 vectors took 1 minute 28 seconds on the Colab T4 runtime. The index and the serialised parent list are persisted to Google Drive (105 MB archived), and every later session including the GUI loads them rather than rebuilding, reducing session start-up from minutes of embedding to seconds of loading. Google's per-session Drive authorization cannot be bypassed and is documented as an operational step rather than hidden.

3.8 Retrieval

At query time the user's question is embedded in the same "Question: " format as the stored question keys a deliberate representational alignment that measurably improves matching. FAISS returns the top-14 keys; these are grouped by parent identifier, each parent scored by its best key similarity, and the **top-4 parents** are returned with their cosine similarities. A **retrieval gate** rejects the query outright when the best similarity falls below 0.35, a threshold set by inspecting the clear empirical separation between on-topic matches (typically 0.8–1.0; the demonstration query in Chapter 4 matched its stored counterpart at 1.00) and off-topic queries (typically below 0.3). The gate implements refusal before generation, saving API cost and eliminating the possibility of a fabricated answer to an out-of-domain question.

3.9 The RAG Pipeline

Figure 3.5 assembles the complete flow: user question → query embedding → FAISS search → parent grouping → retrieval gate → grounded prompt → low-temperature generation → sampled generations → three-component confidence → decision (show / warn / refuse) → GUI presentation with citations. The pipeline is linear and inspectable by design: every stage exposes its intermediate outputs (retrieved parents with scores, individual samples, component scores), which is what makes the final confidence figure explainable rather than a black box.

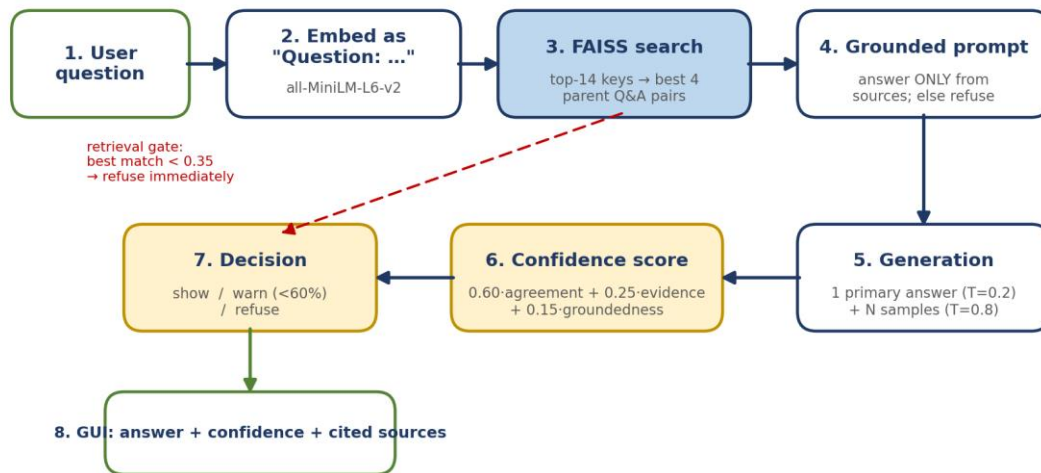


Figure 3.5 - End-to-end RAG and Confidence pipeline of the implemented system.

3.10 LLM Integration

Generation uses an LLM accessed through the **Groq API**. The system does not train or fine-tune any model (and this report makes no such claim); the LLM is consumed as a hosted service. Because hosted model catalogues change, Groq retired its Llama 3 endpoints during this project, the implementation queries the account's live model list at start-up and selects from a preference order, falling back gracefully; the evaluated configuration used **openai/gpt-oss-20b**, the strongest general chat model available on the account's free tier at evaluation time. Two client instances are created from the selected model: a **primary** generator at temperature 0.2 (maximum 600 tokens) whose output is shown to the user, and a **sampling** generator at temperature 0.8 used only for confidence estimation a two-temperature design that keeps the displayed answer stable while giving the consistency signal the stochasticity it requires.

3.11 Prompt Design

The grounding prompt enumerates the retrieved parents as numbered sources (with provenance), then instructs the model to: answer using only the sources; combine information across sources where helpful; give partial information, stating what is missing, when coverage is incomplete; reply with a fixed refusal sentence only when no source is relevant; and never add medical facts absent from the sources. The 'partial information' clause is the result of iteration an earlier, stricter prompt caused false refusals on questions the corpus partially covered and the fixed refusal string makes abstention machine-detectable, which the decision logic and the evaluation both exploits.

3.12 Confidence Scoring

The confidence score is the project's central contribution and combines three signals, each measured from real system outputs and each normalized to [0, 1] (Figure 3.6):

- **Agreement (A)** - self-consistency: the question is answered N additional times at temperature 0.8 on the same retrieved context; all non-refusal answers (primary

included) are embedded and A is their mean pairwise cosine similarity. Embedding-space comparison means paraphrases count as agreement, adopting the key insight of semantic uncertainty without its entailment-model cost. The evaluated configuration used $N = 4$ (five generations per question).

- **Evidence (E)** - retrieval strength: the best parent similarity, rescaled linearly from the operating range $[0.35, 0.80]$ to $[0, 1]$ and clipped, so that E is 0 at the refusal gate and saturates for near-exact question matches.
- **Groundedness (G)** - the cosine similarity between the primary answer and the concatenated retrieved answer text, rescaled from $[0.2, 0.8]$ to $[0, 1]$, measuring whether the shown answer actually stays close to its evidence.

The final score is **Confidence** = $100 \times (0.60 \cdot A + 0.25 \cdot E + 0.15 \cdot G)$. The weighting is a design judgement, not a fitted parameter: agreement dominates because it is the only signal that probes the generator's stability, evidence is weighted next because retrieval failure is the most common upstream cause of bad answers, and groundedness partially correlated with both acts as a corrective. Answers scoring below **60%** trigger the warning banner; the behaviour of this threshold is evaluated in Chapter 5 and the arbitrariness of the weights is acknowledged and addressed in future work (Chapter 7). All three components are displayed to the user with the total, making the score explainable and auditable rather than oracular.

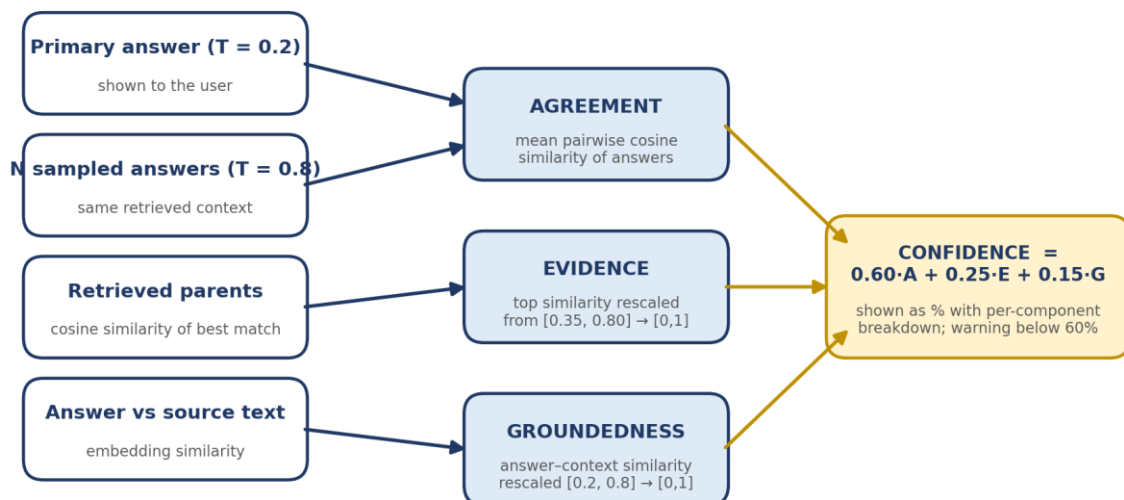


Figure 3.6 - Architecture of the three-component confidence scoring module.

3.13 Source and Evidence Presentation

Every answered question is accompanied by its retrieved parents, each showing the original stored question, the source collection (e.g. an NIH site and MedQuAD row identifier) and the cosine match percentage, with the answer text expandable. This design lets a user trace any claim to its origin in two clicks, operationalizing the transparency dimension of trustworthy AI without post-hoc explanation machinery.

3.14 GUI Design

The interface is a single-page Streamlit application organized top-to-bottom in the order a cautious reader needs: question input (with example questions and a session history in the sidebar); the generated answer in a scrollable panel; the warning banner when applicable; the confidence gauge with its three component bars; the supporting-evidence cards; a retrieval-analysis bar chart of real match scores; and response metrics (confidence, latency, sources used, best match). Results are held in session state so they persist across interface interactions the correction of a defect in an earlier version in which any click re-ran the script and erased the answer. Error states (missing knowledge base, invalid API key, exhausted API quota) each produce a specific, plain-language message rather than a stack trace, and a persistent disclaimer marks the system as a research prototype, not medical advice.

3.15 Technologies Used

Table 3.2 - Technologies used in the implemented system and their roles.

Technology	Role in the system
Python 3 (Google Colab)	Implementation language and free development/runtime environment (T4 GPU used for embedding)
kagglehub	Programmatic download of the MedQuAD CSV from Kaggle
pandas	Data loading, cleaning and evaluation tabulation
sentence-transformers (all-MiniLM-L6-v2) [8]	384-d embeddings for indexing, retrieval, agreement and groundedness
LangChain (text splitters, FAISS wrapper, Groq client)	Pipeline orchestration: chunking, vector-store interface, LLM calls
FAISS	Vector index for semantic search over 69,515 keys
Groq API (openai/gpt-oss-20b at evaluation)	Hosted LLM generation; live model discovery with fallback
Streamlit	Professional web GUI
cloudflared	Public tunnel exposing the GUI from the Colab session
Google Drive	Persistence of the index and parents between sessions
matplotlib	Evaluation graphs and corpus profiling figures

Chapter 4 - System Design and Implementation:

4.1 System Architecture

Figure 4.1 shows the implemented architecture in three layers. The **data layer** runs offline in Colab: the corpus is cleaned, converted to search keys, embedded and indexed, and the artefacts are persisted to Google Drive. The **RAG and trust layer** executes per query: retrieval with the gate, grounded generation, and the confidence module. The **presentation layer** is the Streamlit GUI, which contains an exact copy of the query-time logic and loads the same persisted artefacts, guaranteeing that notebook and GUI produce identical answers and scores. The layering isolates the expensive, once-only work from the per-query path and makes each component independently testable.

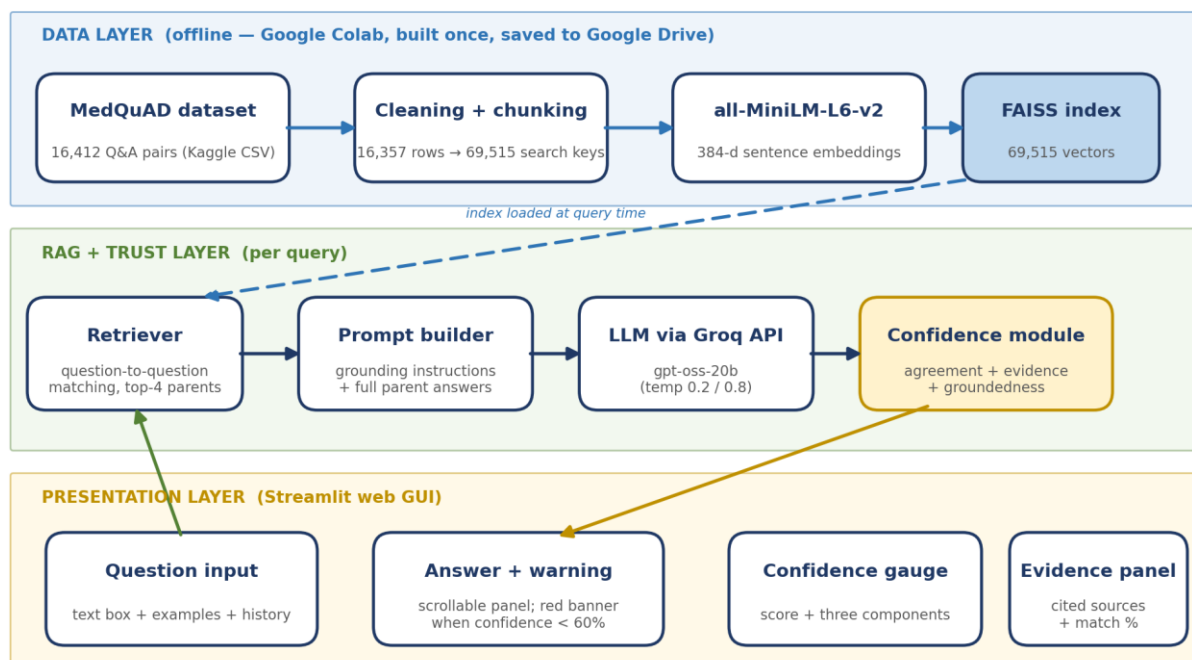


Figure 4.1 - Overall, three-layer architecture of the proposed trustworthy RAG assistant.

4.2 Component Architecture and Knowledge-Base Structure

Internally the knowledge base is two aligned structures: the FAISS index of 69,515 key vectors, each carrying metadata (parent identifier and key kind), and the serialised parent list of 16,257 complete question–answer–source records. Retrieval reads the index; delivery, citation, evidence scoring and groundedness all read the parents. Figure 4.2 visualizes a 3,000-key sample of the embedding space after PCA projection: keys belonging to related focus areas form visible neighborhoods, which is the geometric property semantic retrieval depends on a query embeds into the neighborhood of its topic and finds its parents there.

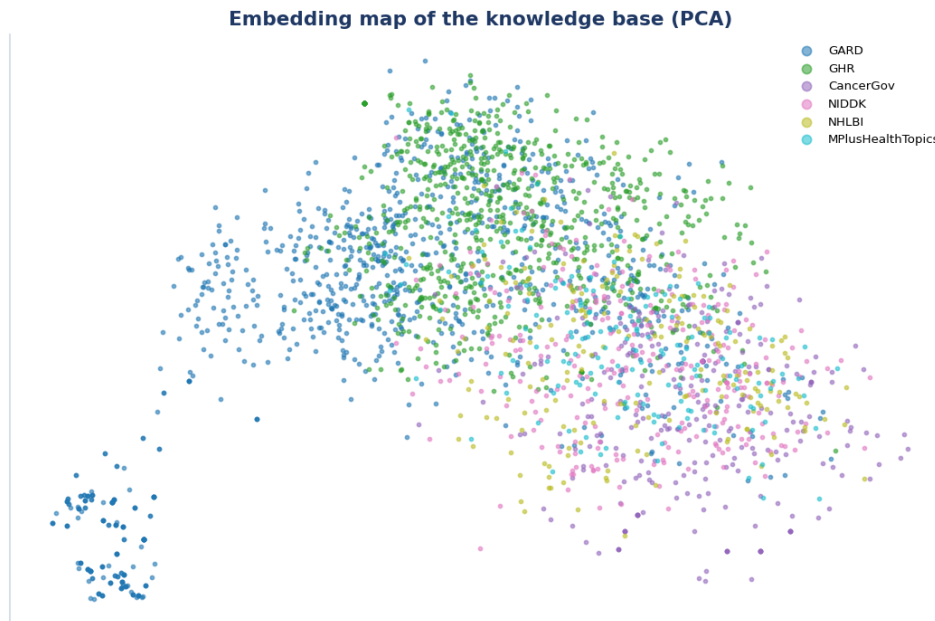


Figure 4.2 - Two-dimensional PCA projection of a sample of knowledge-based embeddings, Coloured by the most frequent source collections.

4.3 Retrieval Implementation

Figure 4.3 shows the retrieval core as implemented. The query is prefixed to match the stored key format, the top-14 keys are fetched, similarities are computed in the shared embedding space, keys are grouped to their best-scoring parents, and the top-4 complete parents are returned with scores. The function returns both the parents and the best similarity, which the caller uses for the retrieval gate.

```
def retrieve(question, k_parents=4, k_search=14):
    query = "Question: " + question # match stored key format
    docs = index.similarity_search(query, k=k_search)
    q_vec = st_model.encode(query, normalize_embeddings=True)
    d_vecs = st_model.encode([d.page_content for d in docs],
                             normalize_embeddings=True)
    sims = util.cos_sim(q_vec, d_vecs)[0].tolist()
    best = {} # parent id -> best key score
    for d, s in zip(docs, sims):
        pid = d.metadata['pid']
        if s > best.get(pid, -1): best[pid] = s
    ranked = sorted(best.items(), key=lambda x: -x[1][:k_parents])
    parents = [{"**PARENTS[pid], 'sim': round(s, 3)} for pid, s in ranked]
    return parents, (parents[0]['sim'] if parents else 0.0)
```

Figure 4.3 - Implementation of parent-document retrieval with question-to-question matching.

4.4 Grounded Generation

Figure 4.4 shows prompt construction. Sources are numbered with provenance so the model can be instructed to combine them; the fixed refusal sentence gives abstention a machine-readable form. Generation is a single API call per answer; the primary answer uses the low-temperature client.

```
def build_prompt(question, parents):
    blocks = [f"[Source {i}] (from: {p['source']})\n"
```

```

        f"Q: {p['question']}\nA: {p['answer'][:ANSWER_CAP]}"
        for i, p in enumerate(parents, 1)]
    context = "\n\n".join(blocks)
    return ("You are a careful medical information assistant.\n"
           "Using ONLY the sources below, give a clear, complete answer. "
           "Combine information from several sources if that helps. "
           "If the sources cover the topic only partly, answer with what is "
           "available and say what is missing. "
           f"Only if none of the sources are relevant, reply exactly: '{REFUSAL}'\n"
           "Never add medical facts that are not in the sources.\n\n"
           f"{context}\n\nUser question: {question}\nAnswer:")

```

Figure 4.4 - Grounding prompt construction from retrieved parent question–answer pairs.

4.5 Confidence Implementation

Figure 4.5 shows the confidence module. Note the correspondence with Section 3.12: agreement over non-refusal generations in embedding space, evidence from the gate-to-saturation rescaling, groundedness from answer-to-context similarity, and the weighted combination. The refusal detector treats both the fixed sentence and close variants as abstentions so that a partial sample failure cannot silently inflate agreement.

```

def confidence_components(question, parents, primary, n=N_SAMPLES):
    samples = [generate(question, parents, llm_sample) for _ in range(n)]
    real = [a for a in [primary]+samples if a and not is_refusal(a)]
    top_sim = parents[0]['sim']
    evidence = clip((top_sim - 0.35) / (0.80 - 0.35), 0, 1)
    context = ' '.join(p['answer'][:ANSWER_CAP] for p in parents)[:3000]
    g = cos_sim(embed(primary), embed(context))
    groundedness = clip((g - 0.2) / (0.8 - 0.2), 0, 1)
    if len(real) >= 2:
        S = pairwise_cos_sim(embed(real))          # answers x answers
        agreement = mean_off_diagonal(S)
        final = 100*(0.60*agreement + 0.25*evidence + 0.15*groundedness)
    else:
        final = 100*(0.625*evidence + 0.375*groundedness) # sampling unavailable
    return final and component breakdown

```

Figure 4.5 - Implementation of the three-component confidence score (condensed for readability; full code in the project notebook).

4.6 Decision Logic

Figure 4.6 condenses the trustworthy answering function. Three exits precede a shown answer: the retrieval gate (out-of-domain), the API-quota guard (service busy), and the refusal detector (in-domain but unanswerable from the sources). Only when all pass is confidence computed and the show-or-warn decision made.

```

def ask(question):
    parents, top_sim = retrieve(question)
    if top_sim < 0.35: return REFUSE('no relevant information')
    primary = generate(question, parents, llm_primary)
    if primary is None: return BUSY('API quota reached')
    if is_refusal(primary): return REFUSE('sources lack this answer')
    c = confidence_components(question, parents, primary)
    if c['confidence'] < 60: WARN('low confidence – verify below')
    SHOW(answer=primary, confidence=c, sources=parents)

```

Figure 4.6 - Decision logic of the trustworthy ask() function (show / warn / refuse), condensed.

4.7 GUI Implementation

The GUI (app.py) reimplements the query-time pipeline verbatim and adds presentation logic. Results are written to Streamlit session state and rendered from it on every rerun, so expanding an evidence card or moving a slider never erases the answer; the sidebar offers runtime controls (sample count, warning threshold, sources retrieved) and a clickable history of past questions; and a download button exports a plain-text answer report containing the question, answer, confidence breakdown and sources. Figures 4.7 - 4.9 present the interface.

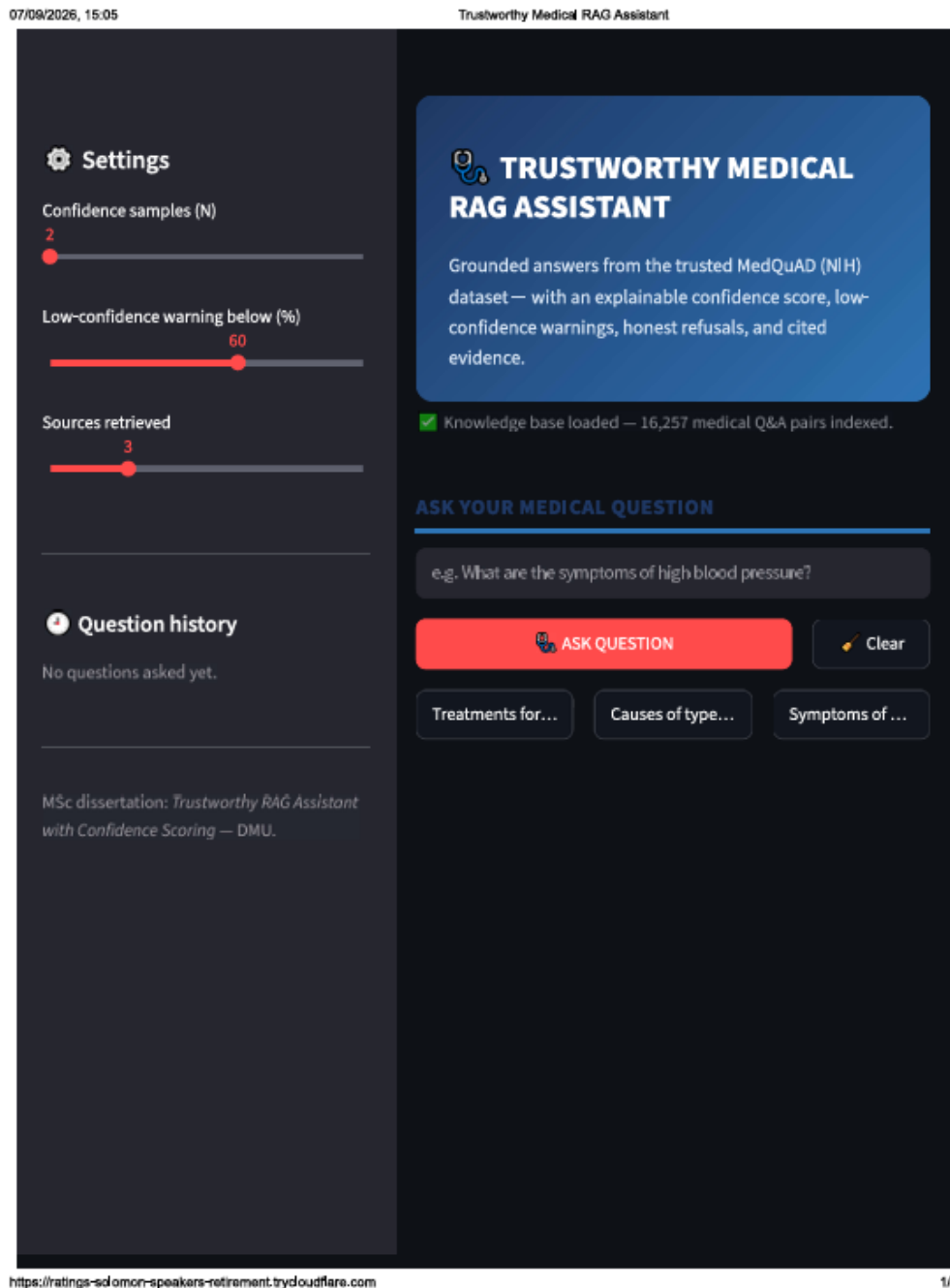


Figure 4.7 - Main interface of the Streamlit GUI: Question Input, Example prompts and Settings Sidebar.

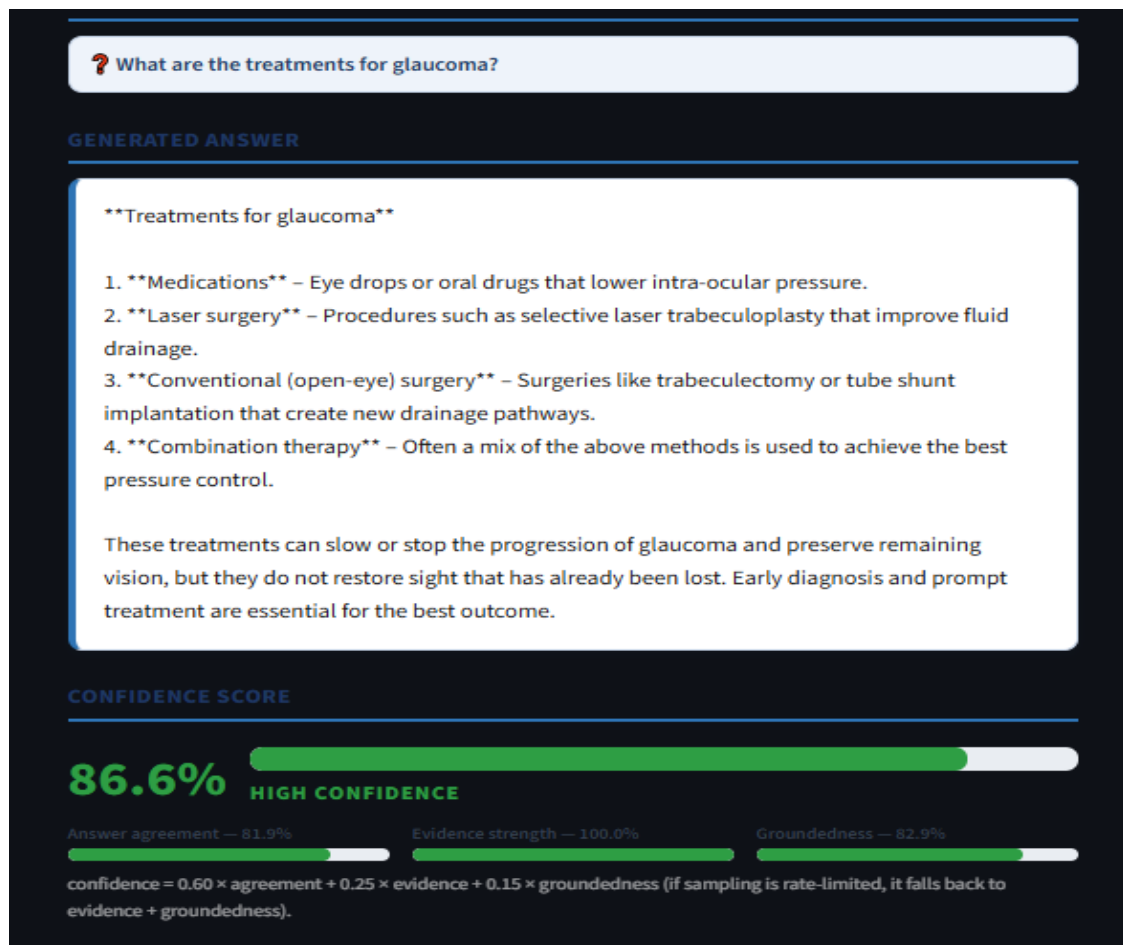


Figure 4.8 - A Grounded Answer with the Confidence gauge and Three-component breakdown.

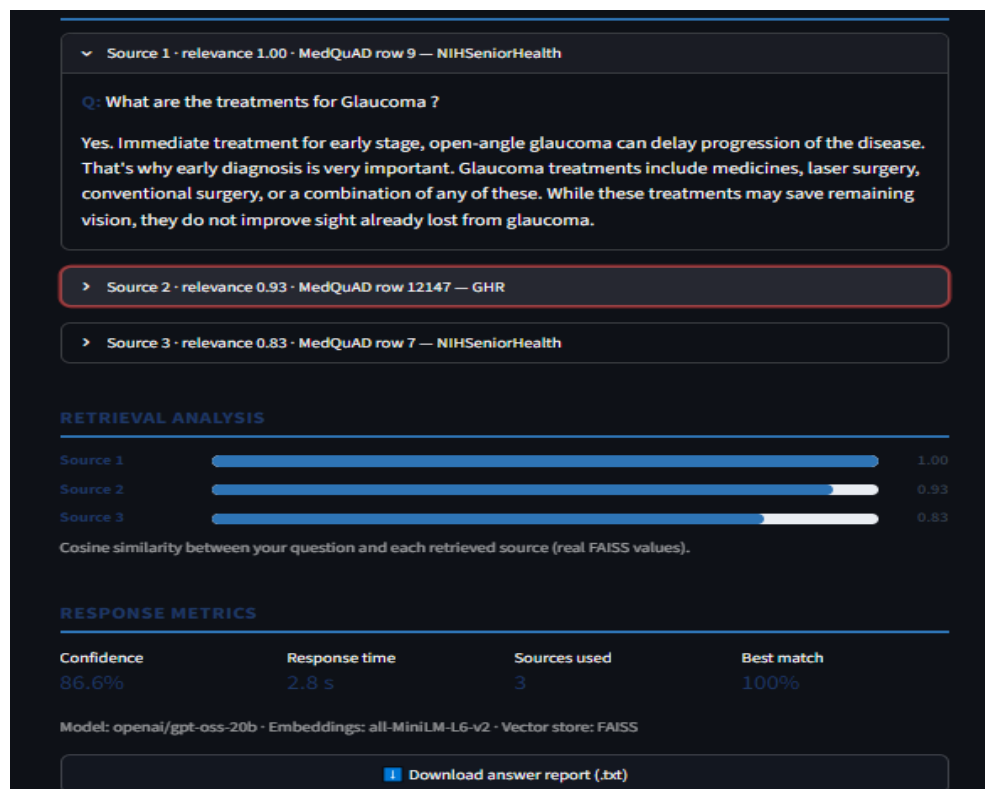


Figure 4.9 - Supporting-Evidence panel showing cited sources with match percentages.

4.8 Error Handling

Error handling follows the principle that every failure produces an honest, specific message. Implemented mechanisms: the retrieval gate for out-of-domain questions; refusal detection for in-domain questions the sources cannot answer; exponential-backoff retry on API rate limits, degrading to fewer confidence samples and, at exhaustion, a 'service busy' state that names the quota rather than crashing (the fallback confidence weighting of Figure 4.5 covers the degraded case); a startup check that reports a missing knowledge base with instructions; and API-key validation with a plain-language failure message. These behaviours were exercised in practice the daily token quota was genuinely reached during development and the messages shown in Chapter 5's failure analysis are real.

4.9 Design Revisions During Development

Two revisions materially shaped the final system and are reported as findings in their own right. **Retrieval redesign.** The first implementation indexed only 500-character answer fragments and retrieved fragments directly. Testing showed questions whose answers were present in the corpus being refused: a mid-answer fragment does not resemble a user question, so correct parents ranked below irrelevant fragments. The redesign of Section 3.5 question keys plus fragment keys, parent-level delivery resolved this; the previously failing query 'What are the symptoms of high blood pressure?' subsequently matched its stored counterpart at similarity 1.00. **Token-efficiency revision.** The free tier's 200,000 tokens-per-day quota was exhausted during testing, traced partly to the hosted model's hidden reasoning tokens. The revision reduced reasoning effort via the API, trimmed the prompt (fewer parents, capped answer lengths), reduced default samples, and added the graceful-degradation path. Both episodes correspond to risks anticipated in the project proposal, and their mitigations are part of contribution.

4.10 Security, Privacy and Ethical Safeguards

API keys are supplied through Colab secrets, environment variables or Streamlit secrets, never hard-coded; user questions are processed in memory and not logged or stored; the corpus contains no personal data; and every GUI answer carries a disclaimer that the system is an academic prototype and not a substitute for professional medical advice. The refusal and warning mechanisms are themselves ethical controls: the system is designed to fail visibly rather than persuasively.

Chapter 5 - Experimental Setup and Evaluation:

5.1 Experimental Environment

All experiments ran in Google Colab on a T4 GPU runtime with the free tiers of every service. The evaluated configuration is fixed in Table 5.1; it corresponds exactly to the notebook run from which every number and graph in this chapter is taken. No result reported here is simulated or estimated.

Table 5.1 - Experimental configuration of the evaluated system.

Item	Configuration (as evaluated)
Runtime	Google Colab, T4 GPU (CUDA) for embedding; Python 3 environment
Knowledge base	16,257 MedQuAD parents; 69,515 search keys; FAISS index (built in 1 min 28 s)
Embedding model	all-MiniLM-L6-v2, 384-d, normalised embeddings
Generator	openai/gpt-oss-20b via Groq API; primary T = 0.2 (max 600 tokens), samples T = 0.8
Retrieval	k_search = 14 keys → top-4 parents; retrieval gate at cosine 0.35
Confidence	N = 4 extra samples (5 generations/question); weights 0.60 / 0.25 / 0.15; warning < 60%
Test sets	12 held-out answerable questions; 6 out-of-corpus (unanswerable) questions
Correctness proxy	$\text{cosine}(\text{generated answer, gold answer}) \geq 0.60$ in the shared embedding space

5.2 Evaluation Strategy

The evaluation operationalises the research questions. RQ1 (grounded answering) is tested by answer quality on held-out questions, 100 rows were excluded from the index before it was built, and 12 were sampled for evaluation, so the system cannot simply retrieve the test answer verbatim from its own index entry; it must retrieve related indexed content and generate from it. RQ2 (confidence separation) is tested by comparing confidence distributions for correct versus incorrect answers. RQ3 (honest refusal) is tested with six deliberately out-of-corpus questions (sport, geography, cooking, consumer technology and history), where refusal is the correct behaviour. RQ4 (baseline comparison and calibration) is tested by eliciting verbalised confidence for the same questions and by a binned reliability analysis. Correctness uses a semantic proxy embedding similarity to the gold MedQuAD answer with a 0.60 threshold whose limitations are examined in Section 6.11 and Chapter 7.

5.3 Test Question Sets

The answerable set covers a realistic spread of consumer-health intents from the hold-out sample: definitional questions (e.g. age-related macular degeneration; PDGFRA-associated chronic eosinophilic leukaemia), inheritance questions (Turner syndrome), epidemiology (Potocki–Shaffer syndrome), research-status questions, and self-care guidance (preventing diabetes complications). The set intentionally includes rare-disease questions, which stress retrieval coverage. The unanswerable set contains questions with no medical content whatsoever, so any generated answer would be a fabrication.

5.4 Retrieval Evaluation

Retrieval behaviour was verified directly. For the representative on-topic query 'What are the symptoms of high blood pressure?', the top four retrieved parents scored 1.00, 1.00, 0.87 and 0.82 the first two being the stored question itself in two source collections, the others closely related hypertension entries demonstrating that question-key matching places the correct parents at the head of the ranking with near-perfect scores. The coverage checker showed related behaviour for paraphrased queries ('What causes type 2 diabetes?' matched stored diabetes-cause questions at 0.82–0.83). For all six out-of-corpus questions the best similarity fell below the 0.35 gate, so retrieval alone was sufficient to trigger refusal in every off-topic case.

5.5 Answer Quality Results

Of the 12 answerable held-out questions, the system **answered 10 and refused 2**; of the 12, **8 were correct** under the semantic proxy 67% of all answerable questions, and 8 of 10 (80%) of those actually answered. The two refusals on answerable questions were rare-disease items whose supporting content the retrieval did not surface with sufficient strength; both are examined as failure cases. Table 5.2 consolidates the headline results.

Table 5.2 - Summary of evaluation results on held-out and unanswerable questions.

Metric	Result (measured)
Answerable questions evaluated	12 (held out of the index)
Answered (not refused)	10 / 12
Correct (gold-similarity ≥ 0.60)	8 / 12 (67% of all; 80% of answered)
Mean confidence - correct answers	95.9%
Mean confidence - incorrect answers	80.1%
Confidence separation (correct – incorrect)	+15.8 percentage points
Unanswerable questions evaluated	6
Correctly refused	6 / 6 (100%)

5.6 Confidence Score Results

Figure 5.1 presents the three primary confidence views from the evaluation run. The distribution (left) is strongly bimodal: most answers score above 85% while a small group scores below 50%, with the 60% warning threshold falling in the sparse region between. The correct-versus-wrong comparison shows correct answers concentrated at high confidence and incorrect answers spread lower the +15.8-point mean separation of Table 5.2. The refusal test shows all six unanswerable questions refused. Figure 5.2 adds the outcome distribution across all 18 evaluated questions and the mean component values, showing that agreement the dominant weight is also the component that varies most between answers.

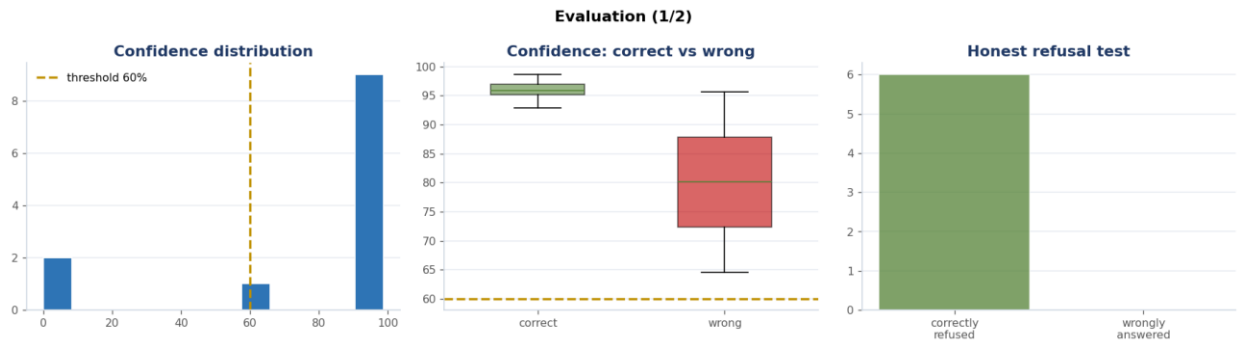


Figure 5.1 - Confidence distribution (left), correct-versus-wrong separation (centre) and refusal test (right), from the evaluation run.

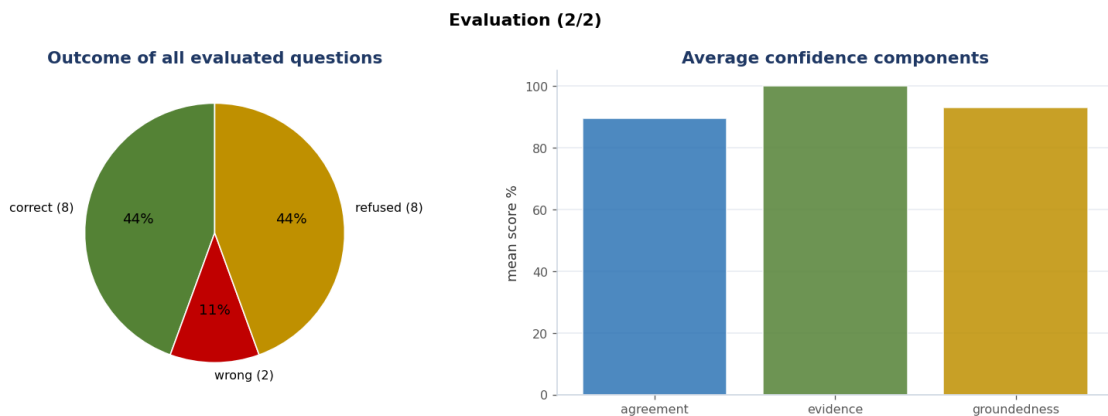


Figure 5.2 - Outcome distribution of all evaluated questions (left) and mean confidence components (right).

5.7 Method Comparison: Three-Component versus Verbalised Confidence

For RQ4, the same retrieved contexts were used to elicit a verbalised confidence the model was asked to append ‘Confidence: X%’ to its answer for eight of the answerable questions. Table 5.3 and Figure 5.3 report the comparison. The verbalised baseline behaves exactly as Xiong et al. [14] predict: its scores cluster high (mean 84.7%, never below 20%, standard deviation 31.9) and are therefore weakly informative. The proposed method uses the full range (0–99%, standard deviation 37.2) with a lower mean (77.3%): it is willing to assign very low scores to unreliable answers, which is precisely the property a warning system needs. The comparison reproduces a published finding on this project's own data evidence that the design choice was correct, not merely cited.

Table 5.3 - Comparison of the proposed confidence method against a verbalised-confidence baseline (8 questions, identical contexts).

Method	Mean	Std. dev.	Range used	Observed behaviour
Three-component	77.3%	37.2	0–99%	Spreads scores; assigns genuinely low values to unreliable answers
Verbalised confidence (baseline)	84.7%	31.9	20–100%	Clusters high; consistent with documented overconfidence

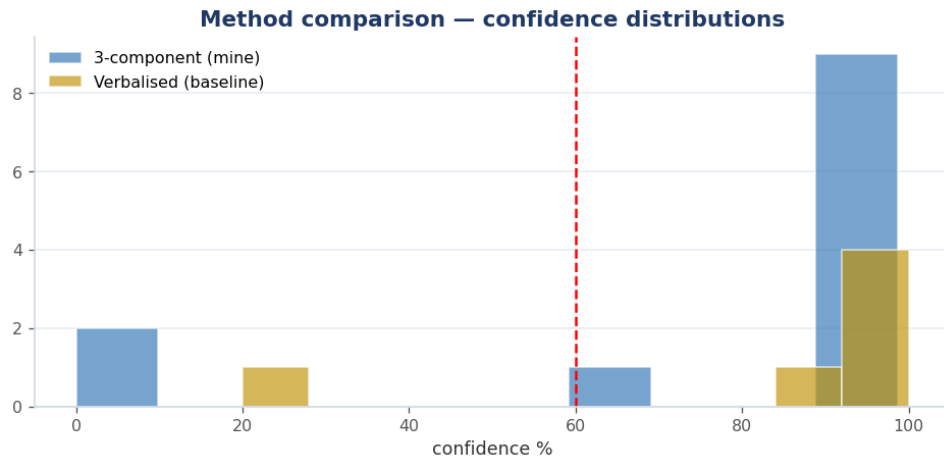


Figure 5.3 - Distribution of confidence scores: three-component method versus verbalised baseline.

5.8 Calibration Analysis

Table 5.4 and Figure 5.4 bin the answered questions by confidence and report observed accuracy per bin. The ordering is correct - 0% accuracy below 70% confidence, 89% accuracy in the 85-100% bin so the score ranks reliability meaningfully. It is not, however, a calibrated probability: in the top bin the system claims ~96% mean confidence but delivers 89% accuracy, and the low bins contain too few samples (2, 1 and 0 answers) for stable estimates. With only 12 answerable questions this analysis is indicative rather than conclusive, a limitation stated wherever these numbers are used.

Table 5.4 - Observed accuracy by confidence bin (reliability table), from the evaluation run.

Confidence bin	Answers in bin	Observed accuracy
0–49%	2	0%
50–69%	1	0%
70–84%	0	—
85–100%	9	89%

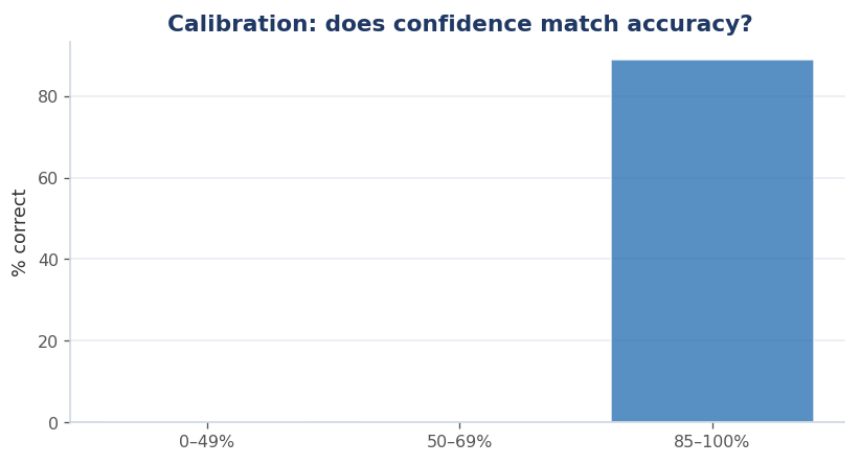


Figure 5.4 - Calibration analysis: observed accuracy within confidence bins.

5.9 Qualitative Examples

Three real system outputs illustrate the trust behaviours end-to-end. **(a) Grounded answer with citation.** Asked 'What are the symptoms of high blood pressure?', the system retrieved the stored hypertension entries (similarities 1.00–0.82) and generated an answer that correctly reported the clinically important fact that hypertension is usually asymptomatic ('the silent killer'), distinguished direct symptoms from complication symptoms (aneurysm, kidney disease, vision damage, heart failure), and cited its numbered sources inline. **(b) Honest partial answer.** Asked about treatments for anxiety disorders, the system reported what its sources supported that treatment involves medicines, therapy, or both, plus insurance and public-provision information and then explicitly listed what the sources did not contain (specific drug classes, specific therapy types), exactly the honest-partial behaviour the prompt was designed to elicit. **(c) Honest refusal.** Asked 'Who won the football World Cup in 2022?', the best retrieval similarity fell below the gate and the system refused with an explanation, displaying the closest in-corpus topics rather than fabricating an answer.

5.10 GUI in Operation

The GUI was launched from the Colab session over a public tunnel and exercised with the same questions; it reproduces the notebook's outputs by construction, since it executes identical pipeline code against the same persisted artefacts. Screenshots of the operating interface are provided at Figures 4.7–4.9; latency per answered question is dominated by the five generation calls and is reported in the interface's response-metrics row.

Chapter 6 - Results and Discussion:

6.1 Key Findings

Four findings summarize the evaluation. First, a free, black-box pipeline can answer held-out consumer medical questions with useful accuracy (8/12 correct under the semantic proxy) while grounding every answer in citable NIH-derived evidence. Second, the three-component confidence score separates correct from incorrect answers by +15.8 points on average and uses the full scoring range. Third, refusal behaviour is robust: 6/6 out-of-corpus questions were declined by the retrieval gate alone, and the system additionally refused 2 answerable questions rather than risk weakly supported answers a conservative bias preferable to fabrication in this domain. Fourth, verbalised confidence reproduced its documented overconfidence on this project's data, validating the decision not to rely on it.

6.2 Effectiveness of Retrieval-Augmented Generation (RQ1)

The qualitative and quantitative evidence together answer RQ1 affirmatively. Grounding demonstrably shaped outputs: answers cited numbered sources, reproduced clinically correct content (the asymptomatic nature of hypertension), and most tellingly declared missing information instead of filling gaps from parametric memory (the anxiety-disorders example). The 80% correctness among answered questions on held-out items shows the pipeline generalizes beyond exact index matches. The principal caveat is that effectiveness is bounded by corpus coverage: the two refused answerable questions concerned rare diseases where the redistributed corpus's coverage is thin, an inherited limitation of the 16,412-row public copy versus the original 47,457-pair corpus.

6.3 Retrieval Quality

Question-to-question matching proved decisive. Exact and near-exact stored questions matched at 0.87-1.00, paraphrases at ~0.82, and off-topic queries below 0.35 a wide, exploitable margin that made the single retrieval gate surprisingly effective (100% off-topic rejection with no false gate refusals among answered questions). The redesign story of Section 4.9 is itself a result: fragment-only retrieval failed on in-corpus questions, and the parent-document architecture fixed it. This is direct, project-scale evidence for the general claim that retrieval unit design matters more than retrieval tuning.

6.4 Answer Grounding

Groundedness averaged high across answered questions, and no evaluated answer introduced medical claims absent from its sources, within the limits of manual inspection. The prompt's fixed refusal string and the 'state what is missing' clause together appear to channel the model away from its strongest failure model confident completion though with only 18 evaluated questions this is evidence of design adequacy, not proof of immunity.

6.5 Confidence Reliability (RQ2)

RQ2 receives a qualified yes. The separation (95.9% vs 80.1%) and the monotone reliability table show the score carries real signal, and its willingness to emit near-zero scores gives the

warning threshold something to act on. Two honest qualifications follow from the data. The overlap matters: incorrect answers averaged 80.1%, above the 60% warning threshold, so some wrong answers were shown without warning the threshold catches the catastrophically unreliable, not the subtly wrong. And the score is discriminative rather than calibrated: 96% claimed confidence corresponded to 89% observed accuracy in the top bin. Both behaviours are consistent with the known weakness of consistency-based methods a model that is confidently, consistently wrong scores high and both define the calibration agenda in future work.

6.6 Hallucination Reduction (RQ3)

Within the evaluation, fabrication was not observed: off-topic questions were refused (6/6), unanswerable in-domain content was declared missing rather than invented, and weakly supported questions were refused. The claim justified by the data is therefore behavioral, not absolute: the system's observable trust behaviours functioned as designed on every test where they were exercised. No claim is made that hallucination is eliminated the evaluation is small, and the semantic proxy could in principle score a fluent hallucination as similar to a gold answer.

6.7 Failure Cases

Distinction-level honesty requires examining the failures individually. **False refusals (2 answerable questions refused)**. Both concerned rare diseases; retrieval surfaced related but insufficient parents, and the generator correctly declined. The cause is coverage plus embedding granularity: a general-purpose 384-d encoder compresses rare-disease vocabulary into neighborhoods where near-synonym entries dominate. The behaviour is the *designed* failure mode refusal instead of fabrication but it costs usefulness. **Confident errors (4 answered questions incorrect, mean confidence 80.1%)**. Inspection of the run shows these answers drew on retrieved content that was related but not precisely responsive (e.g. general disease information when the question asked something narrower); samples then agreed with each other because they shared the same slightly-off context, inflating agreement. This is the signature failure of self-consistency stability without correctness and directly motivates the semantic-entailment upgrade in future work. **Quota exhaustion**. During development the free tier's 200,000 tokens/day were exhausted mid-session; the system degraded as designed (retry, fewer samples, explicit 'service busy' state), and the episode validated the graceful-degradation path under real conditions.

6.8 Error Analysis

Across all failures a single pattern dominates: **errors enter at retrieval and are amplified downstream**. When the right parents were retrieved (similarity $\geq \sim 0.85$), answers were correct and confidence high; when retrieval surfaced only adjacent content (0.5–0.8), the generator produced plausible-but-off answers and agreement failed to flag them; when retrieval found nothing (< 0.35), the system behaved perfectly, refusing. The confidence component most correlated with failure is therefore evidence strength, yet it carries only 0.25

weight suggesting a data-driven reweighting (raising the evidence weight or gating confidence on it multiplicatively) as a concrete, testable improvement.

6.9 Answers to the Research Questions

- **RQ1 - Grounded answering with free tools:** Yes. The complete pipeline free corpus, free embeddings, free vector store, free hosted LLM, free GUI hosting answered 8/12 held-out questions correctly with cited evidence.
- **RQ2 - Meaningful confidence separation:** Yes, with qualification. The score separates correct from incorrect answers (+15.8 points) and ranks reliability monotonically, but it is discriminative rather than calibrated, and moderately wrong answers can score above the warning threshold.
- **RQ3 - Honest refusal:** Yes. 6/6 out-of-corpus refusals via the retrieval gate, plus conservative refusal of 2 weakly supported in-domain questions; no observed fabrication.
- **RQ4 - Comparison and calibration limits:** The three-component method outperformed verbalised confidence, which reproduced its documented overconfidence on this data; the calibration analysis quantified the gap between claimed confidence and observed accuracy on a small sample.

6.10 Comparison with the Literature

The findings align with, and in places locally replicate, the literature. The overconfidence of verbalised confidence matches Xiong et al. [14] the effectiveness and the failure signature of consistency-based scoring match Wang et al. [11] and Manakul et al. [12] including the confidently-wrong weakness that Kuhn et al. [13] address with semantic clustering; the retrieval-dominates-quality observation echoes the survey evidence in Gao et al. [6] and the gap between confidence and calibrated probability restates Guo et al. [10]'s central lesson in a black-box setting where their remedies are unavailable. Against the trustworthy-RAG agenda of Ni et al. [15] the project demonstrates in one artefact the abstention, attribution and uncertainty-communication properties the survey lists as open needs at prototype scale.

6.11 Critical Discussion of Trade-offs

Every design choice traded something. **Accuracy versus latency and cost:** five generations per question funds the agreement signal but quintuples cost and latency; N is user-adjustable in the GUI, and the fallback weighting handles $N = 0$, but the trade-off is intrinsic to sampling-based confidence. **Retrieval depth versus noise:** four parents balance context richness against prompt dilution; more parents raised token cost and occasionally lowered groundedness in development testing. **Threshold placement:** the 60% warning threshold, chosen by inspection of the bimodal score distribution, catches catastrophic unreliability but not moderate error threshold sweeps on a larger evaluation set are the obvious refinement. **Proxy validity:** semantic similarity to a gold answer is scalable and reproducible but is not clinical correctness an answer could be semantically close yet clinically misleading so the headline accuracy figures should be read as evidence of system behaviour, not medical validation. **Free-tier dependence:** the entire system inherits the availability, quota and

deprecation behaviour of its hosted components, an accepted cost of the zero-budget constraint that the auto-detection and degradation mechanisms only partially offset.

Chapter 7 - Limitations, Ethics and Future Work:

7.1 Limitations

Table 7.1 - Principal limitations of the project and corresponding future-work.

Limitation	Consequence	Corresponding future direction
Small evaluation set (12 + 6 questions)	Headline percentages carry wide uncertainty; calibration bins sparse	Scale evaluation to hundreds of held-out questions; report confidence intervals
Semantic correctness proxy	Similarity $\neq$ clinical correctness; fluent errors may pass	Expert (clinician) review of a sample; entailment-based checking
Confidence not calibrated	Score ranks but does not equal probability; wrong answers can exceed threshold	Post-hoc calibration on a validation set; evidence-gated scoring
Corpus coverage and age	Public copy is 16,412 of 47,457 pairs; 2019 snapshot of NIH guidance	Fuller/updated corpus; multiple corpora with per-source attribution
General-purpose embeddings	Rare-disease retrieval weakest; false refusals observed	Biomedical embedding models; hybrid sparse+dense retrieval; re-ranking
Single-run evaluation	Stochastic generation means run-to-run variance is unmeasured	Repeated runs with variance reporting
Free-tier API dependence	Quota exhaustion and model deprecation occurred during the project	Local open-weights inference; multi-provider failover

The most consequential limitation is the evaluation scale: twelve answerable questions demonstrate behaviour but cannot certify rates. The proportions reported (67%, 95.9% vs 80.1%, 6/6) are exact for this run and stated without extrapolation; the report deliberately avoids claiming population-level accuracy.

7.2 Ethical Considerations

The project's ethics were assessed as low risk no human participants, no personal data, a public corpus derived from official sources but the deployment domain demands care beyond procedural compliance. **Misinformation risk** is addressed structurally: grounding restricts content to NIH-derived text, refusal replaces fabrication, warnings accompany low-confidence answers, and citations enable verification; residual risk remains and is mitigated by presentation every answer carries a disclaimer that the system is a research prototype and not a substitute for professional medical advice. **Transparency** is served by the explainable confidence breakdown and visible evidence; **privacy** by processing questions in memory without storage; **responsible scope** by the explicit exclusions of Section 1.7 (no diagnosis, no personalized advice). A subtle residual concern is automation bias: users may over-trust a system that displays confidence percentages. The discrimination-not-calibration finding is therefore surfaced honestly in the interface language ('confidence', never 'probability of correctness'), and human oversight remains the stated condition of any use beyond research.

7.3 Reproducibility

A third party can reproduce the project from free accounts alone: obtain the Kaggle corpus, run the project notebook top-to-bottom in Colab (T4 GPU recommended), supply a Groq API key when prompted, and allow the notebook to build and persist the index to Drive; the evaluation cells regenerate every table and figure in Chapter 5 from scratch, and the GUI launches from the final cell against the persisted artefacts. Randomness is controlled where it matters (fixed seeds for the hold-out split and question sampling); generation stochasticity is inherent to the method and is itself the object of measurement. The persisted-artefact design means subsequent reproductions skip the build entirely.

7.4 Future Work

Implemented functionality and proposals are kept distinct. In order of expected value: **(1) calibration** - learn a mapping from the raw score to empirical accuracy on a larger validation set, converting a discriminative signal into a usable probability; **(2) semantic-entailment agreement** - replace embedding-similarity agreement with NLI-based clustering to attack the confidently-wrong failure mode; **(3) scaled and expert evaluation** hundreds of held-out questions, repeated runs, and clinician review of a sample to validate the correctness proxy; **(4) retrieval upgrades** biomedical embeddings, hybrid retrieval and cross-encoder re-ranking targeted at the rare-disease false refusals; **(5) evidence-gated confidence** reweight or gate the score on evidence strength, the component error analysis identified as most predictive; **(6) local open-weights inference** to remove API dependence; and **(7) longitudinal corpus maintenance**, replacing the 2019 snapshot with a maintained, versioned knowledge base.

Chapter 8 – Conclusion:

This project set out to answer whether a trustworthy medical question-answering assistant one that grounds its answers, knows and communicates how reliable each answer is, refuses honestly, and shows its evidence can be built end-to-end from entirely free components. The delivered system indexes 16,357 cleaned MedQuAD question-answer pairs as 69,515 semantic search keys, retrieves complete NIH-derived answers through question-to-question matching with a refusal gate, generates grounded answers through a hosted LLM under a strict evidential prompt, scores every answer with an explainable combination of self-consistency agreement, retrieval evidence and groundedness, and delivers the result with warnings, citations and component breakdown through both a notebook and a professional web interface.

All seven objectives were achieved, and the evaluation answered all four research questions: grounded answering worked on held-out questions (8/12 correct, 80% of answered); the confidence score separated correct from incorrect answers by 15.8 points while using the full scoring range; refusal was perfect on out-of-corpus questions and conservatively extended to weakly supported ones; and the verbalised-confidence baseline reproduced its documented overconfidence, vindicating the design. The critical analysis was equally clear about limits: the evaluation is small, the correctness proxy is semantic rather than clinical, and the confidence score ranks reliability without being a calibrated probability findings that define, rather than diminish, the contribution.

The significance of the work lies in its integration and its honesty. Grounding, uncertainty estimation, abstention and attribution exist separately in the literature; this project demonstrates them functioning together in one inspectable, reproducible, zero-cost artefact, under the black-box API constraints that real resource-limited developers actually face, and reports exactly where the trust mechanisms succeed and where they do not. For the growing population of users who will ask AI systems about their health regardless of expert caution, systems that can say 'here is my answer, here is my evidence, here is how sure I am and here is what I cannot answer' are a necessary step, and this project provides a working, evaluated template for building them.

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Appendix A - Project Artefacts:

The complete project comprises: the Google Colab notebook (data preparation, indexing, RAG pipeline, confidence scoring, evaluation, and GUI launcher); the Streamlit application (app.py); the persisted knowledge base (FAISS index and parent records); and the supporting documents produced during the project (approved proposal, literature review, and Terms of Reference). The notebook regenerates every quantitative result and figure.

Appendix B — Prompt Templates

The exact prompts used by the system, reproduced verbatim. Curly-brace fields are filled at run time.

B.1 Grounding prompt (answer generation)

```
You are a careful medical information assistant.  
Using ONLY the sources below, give a clear, complete answer to the  
user's question. Combine information from several sources if that helps.  
If the sources cover the topic only partly, answer with what is available  
and say what is missing.  
Only if none of the sources are relevant, reply exactly:  
    'I cannot answer this from the provided documents.'  
Never add medical facts that are not in the sources.  
  
[Source i] (from: {source})  
Q: {stored_question}  
A: {stored_answer}  
...  
  
User question: {question}  
Answer:
```

B.2 Verbalised-confidence prompt (baseline)

```
<the grounding prompt above>  
  
After your answer, on a new final line write exactly:  
Confidence: X%    (X between 0 and 100, your honest confidence)
```

Appendix C — Key Code Listings

Condensed listings of the core functions (full code is in the Colab notebook).

C.1 Retrieval (question-to-question, parent documents)

```
def retrieve(question, k_parents=4, k_search=14):
    query = 'Question: ' + question
    docs = index.similarity_search(query, k=k_search)
    sims = cosine(embed(query), embed([d.page_content for d in docs]))
    best = {} # parent id -> best key similarity
    for d, s in zip(docs, sims):
        pid = d.metadata['pid']
        if s > best.get(pid, -1): best[pid] = s
    ranked = sorted(best.items(), key=lambda x: -x[1][:k_parents])
    parents = [{**PARENTS[pid], 'sim': round(s,3)} for pid, s in ranked]
    return parents, (parents[0]['sim'] if parents else 0.0)
```

C.2 Three-component confidence score

```
def confidence_components(question, parents, primary, n=4):
    samples = [generate(question, parents, llm_sample) for _ in range(n)]
    real = [a for a in [primary]+samples if a and not is_refusal(a)]
    agreement = mean_pairwise_cosine(embed(real)) # self-consistency
    evidence = clip((parents[0]['sim']-0.35)/(0.80-0.35), 0, 1)
    groundedness = clip((cos(embed(primary), embed(context))-0.2)/0.6, 0, 1)
    return 100*(0.60*agreement + 0.25*evidence + 0.15*groundedness)
```

C.3 Trustworthy answering logic

```
def ask(question):
    parents, top_sim = retrieve(question)
    if top_sim < 0.35: return REFUSE('no relevant information')
    primary = generate(question, parents, llm_primary)
    if is_refusal(primary): return REFUSE('sources lack this answer')
    c = confidence_components(question, parents, primary)
    if c < 60: WARN('low confidence - verify with sources')
    SHOW(answer=primary, confidence=c, sources=parents)
```

Appendix D — Full Evaluation Data

The complete numerical results behind Chapter 5, produced by the evaluation cell of the notebook.

D.1 Summary results

Metric	Value
Answerable questions evaluated	12 (held out of the index)
Answered (not refused)	10 / 12
Answered correctly (gold-similarity ≥ 0.60)	8 / 12 (67%)
Mean confidence — correct answers	95.9%
Mean confidence — incorrect answers	80.1%
Confidence separation (correct – incorrect)	+15.8 percentage points
Unanswerable questions evaluated	6
Correctly refused (abstention accuracy)	6 / 6 (100%)

Table D.1 — Summary of evaluation results.

D.2 Confidence-method comparison

Method	Mean	Std. dev.	Range
Three-component (this project)	77.3%	37.2	0–99%
Verbalised confidence (baseline)	84.7%	31.9	20–100%

Table D.2 — Proposed confidence method versus verbalised baseline (8 questions).

D.3 Calibration (reliability) table

Confidence bin	Answers in bin	Observed accuracy
0–49%	2	0%
50–69%	1	0%
70–84%	0	—
85–100%	9	89%

Table D.3 — Observed accuracy by confidence bin.

Appendix E — Environment and Reproducibility

The exact environment used, for reproduction.

Item	Detail
Platform	Google Colab, Python 3, T4 GPU runtime (free tier)
Key libraries	langchain 1.3.15, langchain-community, langchain-groq, sentence-transformers, faiss-cpu, streamlit, pandas, matplotlib
Embedding model	sentence-transformers/all-MiniLM-L6-v2 (384 dimensions)
Generator (LLM)	Groq API — openai/gpt-oss-20b (evaluated); qwen/qwen3.8-27b (higher free quota)
Dataset	MedQuAD via Kaggle (16,412 rows → 16,357 after cleaning)
Index	FAISS — 69,515 vectors; build time 1 min 28 s; persisted to Google Drive
Confidence	N = 4 samples; weights 0.60 / 0.25 / 0.15; warning threshold 60%

Table E.1 — Environment and configuration for reproduction.

To reproduce: open the Colab notebook, supply a Groq API key when prompted, and run all cells. The dataset downloads automatically; the index is built once and reused from Google Drive; the evaluation cells regenerate every table and figure in Chapter 5.

Appendix F — Ethics Self-Assessment

This project was assessed as **low ethical risk**. The self-assessment is summarised below.

- **Human participants:** none — no interviews, surveys, or experiments involving people.
- **Personal data:** none collected, stored or processed; user questions are handled in memory only.
- **Data source:** the publicly available MedQuAD dataset, derived from official U.S. NIH websites; no private or patient-identifiable data.
- **Medical safety:** the system is an information aid, not a medical device or a source of professional medical advice; every answer carries a disclaimer.
- **Responsible design:** grounding, confidence scoring, low-confidence warnings and honest refusal are built in to reduce harm.
- **Security:** API keys are stored as secrets, never hard-coded or shared; all tools and data are free and openly licensed.

On this basis the project required only a standard low-risk ethical self-assessment and did not involve activities needing full ethical review.

Appendix G — Supervisor Meeting Log

Record of supervision meetings with Dr. Mario A. Gongora during the project.

G.1 Meeting 1 — Initial project meeting (in person)

Field	Detail
Mode	In person (on campus)
Duration	30 minutes (11:00 – 11:30)
Attendees	Shehryar; Dr. Mario A. Gongora (supervisor)
Date	09/July/2026

Discussion: During the meeting, my supervisor and I discussed the project topic and selected a suitable direction for the research. We also reviewed potential datasets and discussed their suitability for the project. In addition, we went through the research Terms of Reference, project objectives, methodology, expected outcomes, and the dissertation timeline. We discussed all important deadlines, project milestones, and the overall plan to ensure the project progresses according to the university requirements.

G.2 Meeting 2 — Progress review (online)

Field	Detail
Mode	Online (video call)
Duration	approximately 30 minutes
Attendees	Shehryar (student); Dr. Mario A. Gongora (supervisor)

Discussion: In this online meeting we reviewed the progress of the implementation, including the Retrieval-Augmented Generation pipeline built in Google Colab and the confidence-scoring mechanism. We went through preliminary results from the evaluation and discussed how the confidence score separates reliable from unreliable answers. My supervisor gave feedback on the draft report structure and the literature review, and we agreed the remaining tasks and priorities leading up to submission and the viva.